

Ahsanullah University of Science and Technology



Department of Computer Science and Engineering Spring 2022

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Course Title: Software Development III

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Introduction:

Telemedicine is the use of electronic information and communications technologies to provide and support health care when distance separates the participants. The main goal of telemedicine is to enable remote physicians to see patients virtually whenever and wherever it's needed. With video or audio equipment, medical professionals can meet with patients to diagnose, treat, and perform follow-up care just as any traditional medical visit, but in a virtual setting.

Distance and travel time between patients and care providers can limit access to care. Fortunately, telemedicine can be particularly beneficial for patients in medically underserved communities and those in rural geographical locations where clinician shortages exist. It reduces expenses for patients to transport to another location, and thus also reduce the risk of transmission of contagious viruses and illnesses to others. Moreover, technology used for telemedicine appointments is designed to be secure and it protects patient privacy. Plus, it ensures patients get their doctors' undivided attention

Hence the impact and applications for telemedicine with hospitals, providers, and patients continues to increase as all parties continue to adapt to receiving and delivering care in this modern healthcare setting.

Motivation:

In remote and rural areas, patients have to travel long distances to see a specialist. Moreover, people do not want to go through the hassle of waiting in long queues on their medical visit. Plus, taking a day off from work for a healthcare checkup seems to be troublesome for many. Besides, during emergency and at midnight, getting access to doctors can be difficult.

All the above problems can be solved if there is an application where one can easily connect to a specialist from the comfort of one's home. This has motivated us in making "DigiCure". "DigiCure" proposes to make telemedicine simple and convenient to use for both practitioner and patient.

"DigiCure"

App Specifications:

Database: Firebase

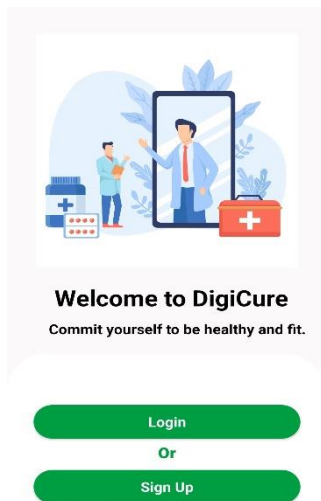
Platform: Android Studio

Language: JAVA

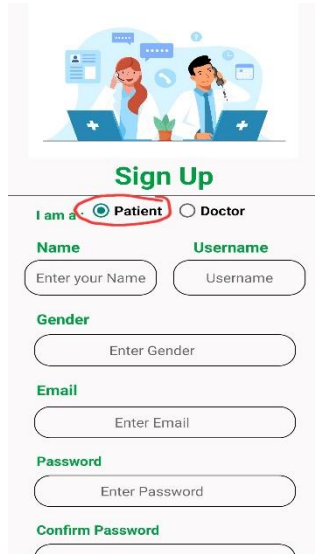
API Level: (API 23: Android 6.0 (Marshmallow))

Video Conference SDK: Jitsi

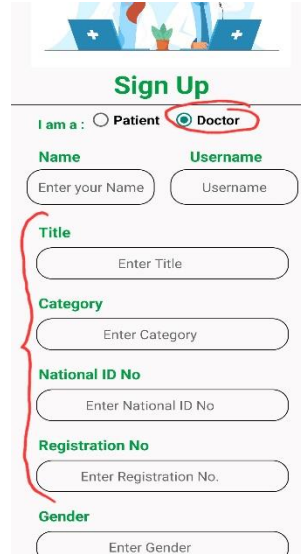
Design Overview:



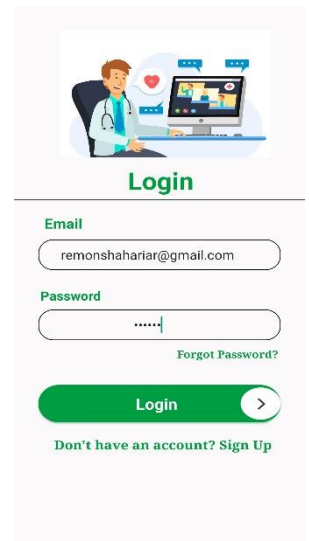
Main Activity



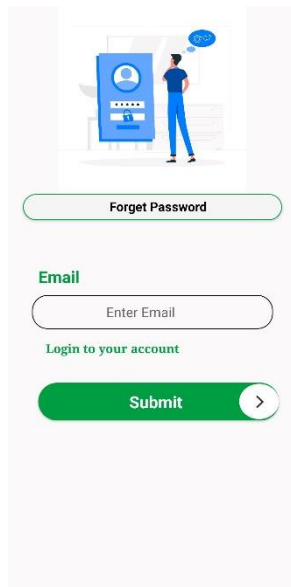
SignUp as Patient



SignUp as Doctor



Login



Forgot Password



Home Fragment



Doctor Fragment



Medicine Fragment

Doctor1

Enter Username

Type : Doctor

Username : doctor97

Title : Dr.

Category : ENT

National ID : 39825566

Registration No: AA232323

Gender : Male

Logout

Profile Fragment

Video Conference

Join Video Conference

Enter Security Code

Enter Code

Join

Video Conference Activity

DigiCure

Video Conference Medicines Profile

Talk with us 0961089679

Top Doctors See All

Dr. Sarwar Ahmed Neurologist 10:00 AM - 1:00 PM APPOINTMENT

Dr. Mubina Akter Gynecologist APPOINTMENT

Night Mode

DigiCure

Authentication

Users Sign-in method Templates Usage Settings

Search by email address, phone number, or user UID Add user

Identifier	Providers	Created ↓	Signed In	User UID
doctor1@gmail.com		Feb 21, 2023	Feb 21, 2023	GpmJI3DSGtauWeE26bfysvUOUw...
remonshaharar@gmail.com		Feb 21, 2023	Feb 21, 2023	cmS31A2tyWbIMgkLMgZVhWhgN...

Authentication

DigiCure

Realtime Database

Data Rules Backups Usage

https://digi-cure-b4bb6-default-rtdb.firebaseio.com

datauser

doctor97

- category: "ENT"
- email: "doctor1@gmail.com"
- fname: "Doctor1"
- gender: "Male"
- lname: "doctor97"
- natioNo: "39825566"
- pass: "123654"
- regiNo: "AA232323"
- title: "Dr."
- type: "Doctor"

Realtime Database

Features:

1. **Registration:** Both doctor and patient can register filling all the necessary information.
2. **Log In:** They can log into their accounts using their username and password to use the app.
3. **Authentication:** A user can reset their password if they forget it through their email address.
4. **Doctor search/selection:** A search bar is available in the "Doctor" page to search for a specific doctor.
5. **Video Conference:** A user can start video conference by giving a specific code to the meeting.
6. **List of all doctors:** List of all doctors can be viewed.
7. **View Profile Information:** User can view their information in their profile.
8. **Night Mode:** Night Mode can be turned on by simply double clicking the Night Mode button.
9. **Navigation Menu:** A user can use the Navigation Menu available on the landing/home page to select which page they want to go.
10. **Available Medicine List:** Available medicine list is given for a user to order.
11. **Log Out:** A user can log out of his/her account.

About Digicure:

"Digicure" is a telemedicine app where both a patient and a doctor can register filling all the required information. While logging in, if they forget their password, they can select the "forgot password" option and then using their email they can set a new password. After logging in, they can view the Home Page where they can see the top doctors' list, navigation menu, video conferencing facility, order medicine etc. Moreover, the user can select night mode view available in the Home Page by simply double clicking it. With the navigation menu, a user can easily select the type of page they want to visit. If the user selects the "Video Conference" button, he/she has to give a specific code to start the meeting and the doctor is also required to use the same code to join the meeting. In the Doctor page, the user can view the different categories of doctors available and book an appointment with them. In the Order Medicine page, they can view the medicines that are available and order through the cell phone number provided on the screen. The user can also view his/her profile to see the required information they have registered with. He/she can easily log out of the account by simply using the log out button available in the profile page.

Difficulties faced while making the application:

1. **Video Conference:** When the join meeting button is clicked, the video chat crashes and ends. It could be a server problem but we were not able to figure out the solution.
2. **Profile Page Data:** We were not able to send the data of a user from the database to view in the profile. But if we search the username, the data can be viewed, but it does not show automatically.
3. **Notifications:** When an appointment is booked by a patient, we could not send that notification to the specific doctor.

Conclusion:

Even though we were not able to provide all the features that we initially hoped to and also solve the errors, we still learned a lot from making our project. Through this course, we can now create basic applications with the help of Android Studio and Firebase database.